Assignment for *"Advanced Statistical Inference" 2026* Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles

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Abstract

This report presents a partial reproduction of *Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles* (1). We reproduce four representative experiments: toy regression uncertainty estimation, MNIST ensemble classification, out-of-distribution uncertainty using predictive entropy, and the effect of ensemble size on predictive performance. Code and figures are available at: `https://github.com/Yassine-Trabelsi10/Simple-and-Scalable-Predictive-Uncertainty-Estimation-using-Deep-Ensembles`.

1 Introduction

Modern neural networks often achieve strong predictive accuracy but can be overconfident. This is problematic in applications where uncertainty matters. The paper by Lakshminarayanan et al. proposes Deep Ensembles, a simple method for estimating predictive uncertainty by training several neural networks independently and averaging their predictions.

This report reproduces a subset of the paper's results using a Colab notebook. The goal is not to reproduce every benchmark from the original paper, but to reproduce the main qualitative conclusions on regression, classification, ensemble size, and out-of-distribution uncertainty.

2 Summary of the Paper

2.1 Motivation

Many probabilistic machine learning methods estimate uncertainty using Bayesian inference. However, Bayesian neural networks often require complex approximate inference methods. Deep Ensembles provide a simpler alternative: train multiple neural networks with different random initializations and combine their predictions.

2.2 Deep Ensembles

For classification, each network predicts class probabilities:

$$p(y \mid x, \theta_m),$$

where θ_m denotes the parameters of model m.

The ensemble prediction is:

$$p(y \mid x) = \frac{1}{M} \sum_{m=1}^{M} p(y \mid x, \theta_m),$$

where M is the number of models.

For regression, each network predicts a mean and variance:

$$\mu(x), \sigma^2(x).$$

This allows the model to represent predictive uncertainty directly. The paper evaluates Deep Ensembles using negative log-likelihood, Brier score, classification accuracy, and out-of-distribution uncertainty.

2.3 Contributions of the Paper

The paper makes three main contributions. First, it demonstrates that ensembles of independently trained neural networks can provide uncertainty estimates that are competitive with more sophisticated Bayesian approaches. Second, it shows that Deep Ensembles achieve strong predictive performance while remaining easy to implement using standard optimization techniques such as stochastic gradient descent. Third, the paper evaluates uncertainty estimation in several challenging settings, including regression, image classification, adversarial examples, and out-of-distribution detection.

An important conclusion of the paper is that accurate uncertainty estimation does not necessarily require approximate Bayesian inference. By combining multiple diverse models, Deep Ensembles can capture predictive uncertainty while avoiding the complexity associated with Bayesian neural networks.

The authors compare Deep Ensembles against several alternative approaches, including MC-Dropout and probabilistic neural networks. Across many experiments, Deep Ensembles achieve competitive or superior performance in terms of predictive accuracy, calibration, and uncertainty estimation quality.

3 Reproduced Experiments

3.1 Toy Regression Uncertainty

We first reproduced a toy regression uncertainty experiment. A probabilistic neural network was trained to predict both mean and variance, and several independently initialized networks were combined into an ensemble.

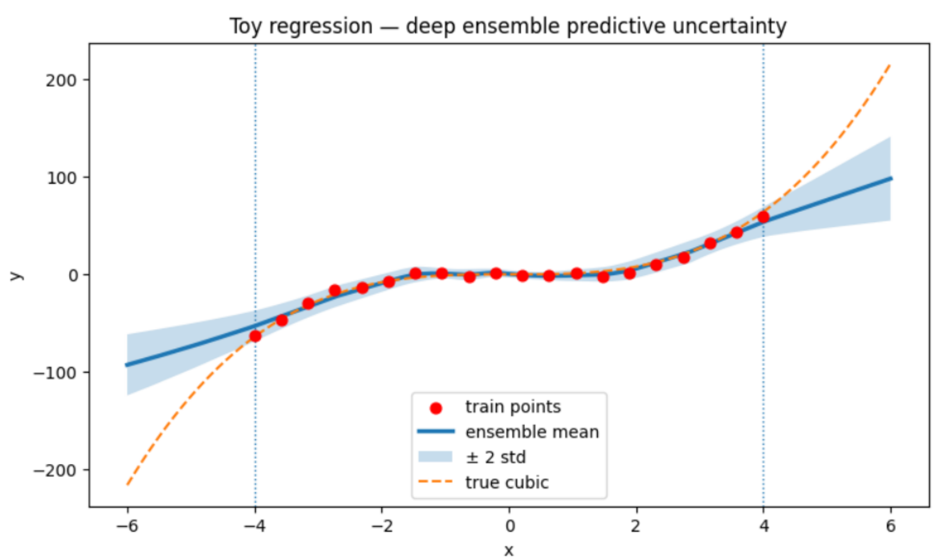

Figure 1: Toy regression experiment reproduced using Deep Ensembles. The predictive uncertainty (shaded area) increases outside the training region, consistent with the behavior reported in the original paper.

The reproduced result shows the expected behavior: uncertainty is lower near observed training data and increases in regions where data are scarce. Inside the training interval, the ensemble predictions closely follow the underlying function and the uncertainty band remains narrow. Outside the observed region, the uncertainty rapidly grows, reflecting the model's lack of knowledge. This behavior is one of the key motivations for uncertainty estimation and matches the qualitative findings reported in the original paper.

3.2 MNIST Deep Ensemble Classification

We trained five independent neural networks on MNIST and combined their predicted probabilities. The ensemble was evaluated using accuracy, negative log-likelihood (NLL), and Brier score.

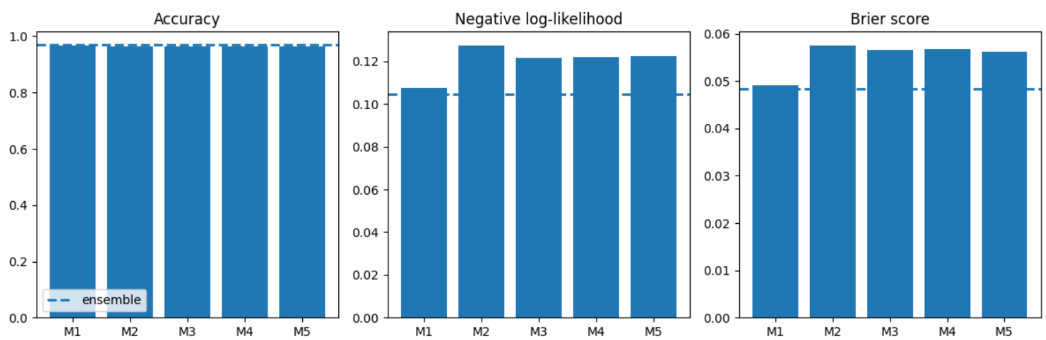

Figure 2: MNIST classification performance of the Deep Ensemble. Accuracy, negative log-likelihood (NLL), and Brier score are reported for individual models and compared with the ensemble prediction.

The ensemble achieved high accuracy and low uncertainty-related losses, showing that the method produces strong predictive performance and meaningful uncertainty estimates. While the individual models already perform well, combining their predictions provides more stable probability estimates and improves the quality of uncertainty measurements. This illustrates the central idea of Deep Ensembles: averaging multiple diverse models can improve both predictive performance and confidence estimation.

3.3 Effect of Ensemble Size

To reproduce the ensemble-size analysis from the paper, we evaluated ensembles with $M = 1$, $M = 3$, and $M = 5$ models.

M	Accuracy	Error	NLL	Brier
1	0.9678	0.0322	0.1075	0.0491
3	0.9698	0.0302	0.1066	0.0490
5	0.9694	0.0306	0.1046	0.0484

Table 1: Effect of ensemble size on MNIST performance.

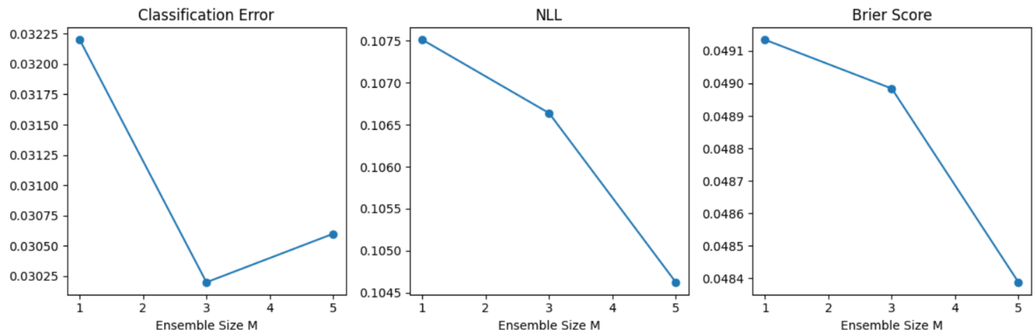

Figure 3: Effect of ensemble size on predictive performance. Increasing the number of ensemble members generally improves uncertainty-related metrics such as negative log-likelihood (NLL) and Brier score, consistent with the observations reported in the original paper.

The NLL and Brier score improve when increasing the ensemble size, which is consistent with the paper's conclusion that larger ensembles improve uncertainty estimation. Although the classification accuracy changes only slightly, the uncertainty-related metrics show a clear improvement. This suggests that larger ensembles mainly enhance the reliability of predicted probabilities rather than dramatically increasing classification accuracy.

3.4 Out-of-Distribution Uncertainty

Finally, we evaluated predictive entropy on MNIST and FashionMNIST. MNIST is the in-distribution dataset, while FashionMNIST is used as an out-of-distribution dataset.

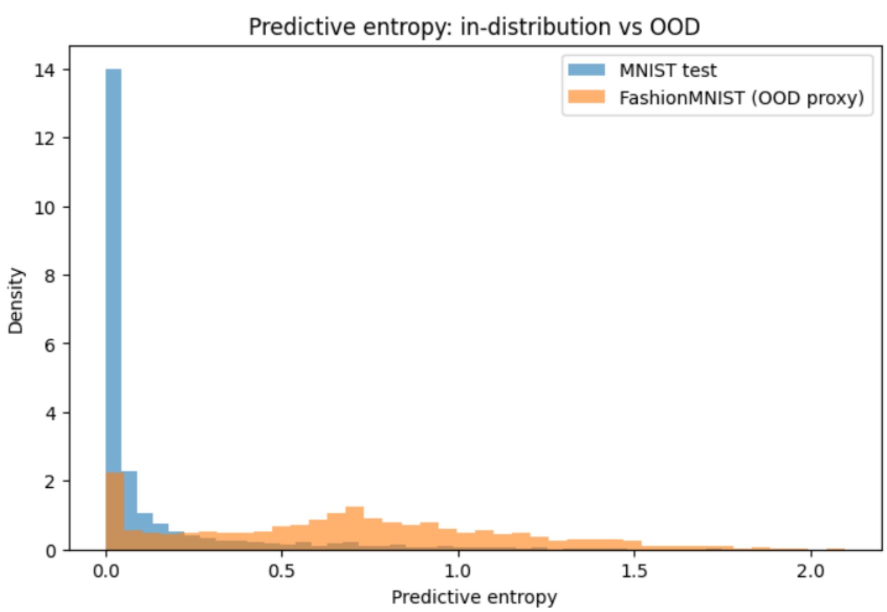

Figure 4: Predictive entropy for in-distribution (MNIST) and out-of-distribution (FashionMNIST) samples. The ensemble produces significantly higher entropy values for unfamiliar inputs, indicating greater predictive uncertainty.

The ensemble assigns higher entropy to out-of-distribution samples, meaning it is more uncertain on unfamiliar inputs. This is a desirable property because a reliable uncertainty estimation method should recognize when it encounters data that differ from the training distribution. The clear separation between the entropy distributions of MNIST and FashionMNIST indicates that the ensemble can distinguish familiar and unfamiliar inputs using predictive uncertainty.

4 Discussion

The reproduced experiments support the main message of the paper: Deep Ensembles are simple to implement and provide useful predictive uncertainty estimates. The toy regression experiment shows uncertainty increasing away from the training data. The MNIST classification experiment confirms that averaging independent models preserves strong predictive accuracy. The ensemble-size experiment shows that uncertainty metrics such as NLL and Brier score improve as more models are added. The entropy experiment shows that the model is more uncertain on out-of-distribution data.

A particularly interesting observation is that the uncertainty estimates remain meaningful even though the method is not explicitly Bayesian. This supports one of the main claims of the paper: practical uncertainty estimation can be achieved through model diversity alone.

The ensemble-size experiment also highlights the trade-off between computational cost and predictive quality. While larger ensembles generally improve uncertainty metrics, they require training and storing multiple models. In practice, the choice of ensemble size depends on available computational resources and application requirements.

The main challenge was computational cost. The original paper includes larger experiments on SVHN and ImageNet, as well as comparisons with MC-Dropout and adversarial training. These experiments were not reproduced because they require more training time and more implementation complexity. Instead, this reproduction focuses on smaller but representative experiments that capture the core conclusions of the paper.

5 Conclusion

This report presented a partial reproduction of the paper *Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles*. Four representative experiments were reproduced: toy regression uncertainty estimation, MNIST ensemble classification, ensemble-size analysis, and out-of-distribution uncertainty detection.

The reproduced results are consistent with the qualitative conclusions reported in the original paper. In particular, uncertainty increases in regions with little training data, larger ensembles generally improve uncertainty metrics, and predictive entropy is higher for unfamiliar inputs.

Overall, Deep Ensembles provide a simple, scalable, and effective alternative to more complex Bayesian approaches for predictive uncertainty estimation. The experiments conducted in this reproduction confirm that meaningful uncertainty estimates can be obtained using standard neural-network training procedures.

References

[1] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles. In *Advances in Neural Information Processing Systems*, 2017.

AI Use Statement

This assignment was completed with the assistance of artificial intelligence tool ChatGPT.

AI tool was used to:

- Explain theoretical concepts related to predictive uncertainty and Deep Ensembles.
- Assist with debugging and improving the implementation in the notebook.
- Suggest improvements to the structure and wording of the report.
- Provide feedback on the interpretation of the experimental results.

All experiments were executed by the author, and all reported results were obtained from the submitted code. The final responsibility for the code, experiments, analysis, and conclusions remains entirely with the author.

The use of AI tool was limited to assistance and guidance. All reported results, figures, interpretations, and conclusions were manually checked to ensure consistency with the original paper and the reproduced experiments.